

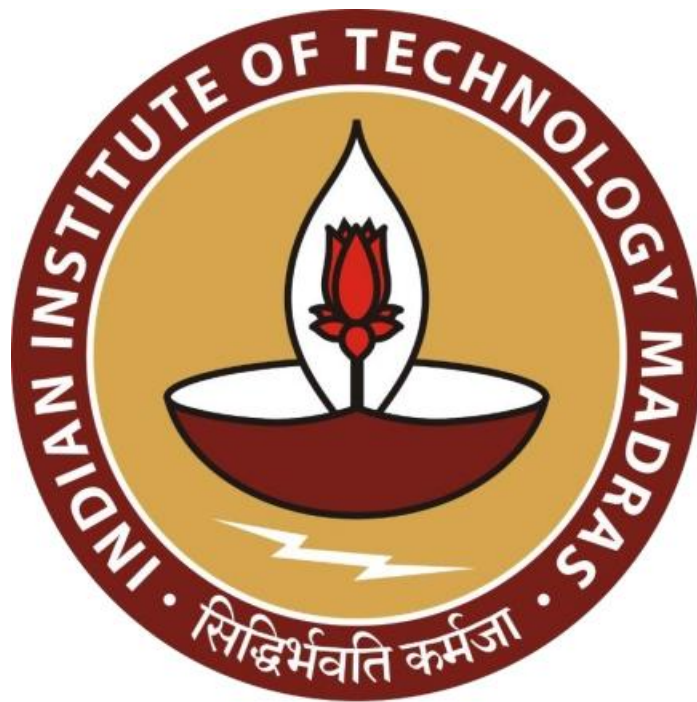
Data-Driven Inventory Planning for a Retail Pharmacy

Final report for the BDM capstone Project

Submitted by

Name: SPANDAN BHOI

Rollnumber : 23f3002227



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai Tamil
Nadu, India, 600036

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1.Executive Summary

Rahul Medical Store is a local pharmacy in Sambalpur, Odisha, dealing in prescription medicines, OTC products, and essential healthcare items. Discussions with the store owner revealed several operational challenges, including frequent expiry of medicines due to unstructured stock monitoring, reliance on experience-based decision-making rather than data-driven forecasting, and occasional sales of non-prescription medicines without proper documentation. These issues indicated the need for a more systematic and transparent approach to inventory management.

This project is based on five months of actual sales and purchase data collected from April to August 2025. The data included details such as product names, batch numbers, quantities, expiry dates, GST, and transaction values. All analysis was carried out using Excel, with the aim of creating a practical and easy-to-use system that a small retail pharmacy can adopt without advanced technical tools.

The analysis shows that the store's highest sales occurred in May (around ₹1.24 lakh), while purchases were highest in April (about ₹1.13 lakh), indicating that most stocking is done early in the cycle. Sales from June to August remained fairly stable, averaging ₹1.1–1.2 lakh per month, which reflects consistent customer demand. The ABC analysis highlighted that 41% of the total inventory value lies in Category A medicines, making them critical from a cost and monitoring perspective. The expiry-risk analysis found that around 23% of the inventory is nearing expiry within the next six months, while the stock-age analysis identified several slow-moving medicines that have remained in stock for more than six months.

Based on these findings, the report recommends adopting staggered purchasing for high-value medicines, prioritizing the clearance of near-expiry and slow-moving stock, and maintaining better documentation for non-prescription sales. Together, these steps can help the store reduce wastage, improve purchase planning, and move towards a more data-driven and efficient inventory management process.

2.Detailed Explanation of Analysis Process

2.1 Data Collection

Data was collected from Rahul Medical Store's physical sales and purchase registers for a period of five months (April–August 2025). The dataset included the date of transaction, product name, batch number, expiry date, quantities, GST details, and total transaction amounts. Both sales and purchase records were manually entered into Excel for analysis.

2.2 Data Cleaning and Pre-Processing

The raw dataset obtained from the sales and purchase registers required several cleaning steps before it could be used for analysis. Since the entries were manually recorded by the store owner, there were a few inconsistencies in the format and structure. The data cleaning process ensured that the dataset became accurate, complete, and suitable for analysis using Excel. This step was essential because even small inconsistencies could lead to incorrect classifications in ABC analysis or wrong identification of near-expiry items.

2.2.1 Correction of date formats:

One of the main issues observed in the raw file was the inconsistency in date formats. Some dates were written in DD/MM/YY format, while others were in text format (e.g., "Apr-25"). To ensure uniformity, all dates were converted into Excel's standard date format using the "Text to Columns" tool and manual correction where required. This ensured accurate month-wise grouping for sales and purchase trend analysis.

2.2.2 Handling missing values:

A few entries, such as thermometers, were missing expiry dates because the store owner did not mention whether the product was digital or mercury-based. Since mercury thermometers do not expire but digital ones do, these records could not be classified correctly in the expiry analysis. After discussing with the store owner, these records were retained for sales and purchase analysis but excluded from expiry-related calculations to avoid misclassification.

2.2.3 Removal of duplicates and blank entries:

During manual verification, a few repeated rows and blank records were found in the purchase sheet. These were removed to avoid double-counting. Blank rows were cleaned using Excel filters and the "Go To Special → Blanks" function.

2.2.4 Standardization of product names:

Product names appeared in multiple formats (e.g., "Multivitamin Syrup", "Multivitamin syp", "Multivitamin"). These were standardized into a single format to ensure correct grouping in Pivot Tables. Without this standardization, the same product would appear repeatedly as separate items in ABC classification or stock-age analysis.

2.2.5 Validation of numerical fields:

Fields such as Quantity, MRP, GST Amount, and Total Amount were checked for non-numeric entries or accidental spaces. Excel's "ISNUMBER" and sorting functions were used

to identify inconsistencies. All monetary fields were corrected to ensure accurate computation of total purchase value and GST.

After completing all these steps, the dataset became clean, consistent, and ready for analysis. This cleaned version was stored as the master dataset and used for all subsequent analyses including ABC classification, expiry risk assessment, and stock-age evaluation.

2.3 Final Data Structure After Pre-Processing

After completing the cleaning process, the dataset was standardized into a structured, analysis-ready format. Both the sales and purchase registers were organized into consistent column headers, ensuring compatibility with Excel analysis tools such as Pivot Tables and charts. The final structure contained the following fields:

- Date
- Product Name
- Batch Number
- Expiry Date
- Quantity
- MRP
- GST (%)
- GST Amount
- Total Amount (₹)
- Expiry Status (Safe / At Risk)
- Stock Age (in months)

This structured format ensured that all analytical methods—ABC classification, stock-age analysis, expiry-risk evaluation, and trend summaries—could be performed accurately without inconsistencies affecting the results.

RAHUL MEDICAL STORE,Sambalpur,Odisha											
Purchase Register											
Period:- April 25 - August25											
Date	Customer/Supplier	Product Name	Batch No	Expiry Date	Quantity	Rate (₹)	Amount (₹)	GST (%)	T Amount	Total (₹)	Expiry Status
2025-04-01	Popular Medical Store	Allegra 120mg	104	Jul-26	10	211.78	1890.89	12	226.91	2117.80	Safe
2025-04-01	Popular Medical Store	Multivitamin Syrup	892	Dec-25	10	131.61	1175.09	12	141.01	1316.10	At Risk
2025-04-01	Taj Pharmsutical	Clotrimazole Cream	725	May-26	25	187.55	4186.38	12	502.37	4688.75	Safe
2025-04-01	Popular Medical Store	Metformin 500mg	115	Apr-26	10	20.68	184.64	12	22.16	206.80	Safe
2025-04-01	Utkal Pharma	Ibuprofen 400mg	395	May-26	10	7.63	68.13	12	8.18	76.30	Safe
2025-04-01	Utkal Pharma	Combiflam	442	Jun-26	50	44.57	1989.73	12	238.77	2228.50	Safe
2025-04-02	Popular Medical Store	Zincovit	734	Mar-26	10	82	732.14	12	87.86	820.00	Safe
2025-04-02	Siba Medical Store	Amlodipine 5mg	600	Jan-26	10	71.31	636.70	12	76.40	713.10	At Risk
2025-04-03	Odisha Chemist & drugist	Volini Gel	994	Jan-26	25	218.36	4874.11	12	584.89	5459.00	At Risk

Table1: Sample of Cleaned Purchase Register Dataset after Pre-processing

2.4 Analysis Methods and Why They Were Used

This section describes how the analysis was performed and *which problems* each method addresses.

2.4.1 Monthly Sales and Purchase Trend (EDA method)

Purpose: To understand stock movement patterns and evaluate whether procurement decisions were aligned with actual customer demand.

Method: Monthly sales and purchase data from April to August 2025 were summarized using Pivot Tables in Excel. Pivot Charts were then used to visualize month-wise trends in sales and purchases.

Explanation: This method helped identify peak sales and purchase months, compare procurement levels against revenue generation, and assess whether inventory was being purchased in excess or in line with demand. The analysis provided insights into procurement timing and stock utilization efficiency.

2.4.2 ABC Classification (Diagnostic Method)

Purpose: To identify high-value medicines that contribute most to total purchase expenditure and require strict monitoring to reduce financial risk.

Method: The total purchase value of each medicine was calculated. Cumulative value percentages were then computed and used to classify medicines into three categories:

- **Category A:** Top 80% of cumulative value
- **Category B:** Next 15% of cumulative value
- **Category C:** Remaining 5% of cumulative value

Explanation: This classification helped prioritize inventory control efforts by highlighting medicines with the highest financial impact. Category A items require close monitoring, while Category B and C items can be managed with relatively lower control intensity.

2.4.3 Expiry Risk Analysis (Diagnostic Method)

Purpose: To reduce losses caused by medicines nearing expiry due to overstocking or slow movement.

Method: A date-based rule was applied using Excel formulas to classify medicines as Safe or At Risk. Medicines expiring between August 2025 and January 2026 were labeled as *At Risk*, while those expiring after January 2026 were classified as *Safe*.

Explanation: This method helped identify medicines that required immediate attention, such as discounting, promotions, or supplier returns, to minimize expiry-related wastage.

2.4.4 Stock Age Analysis (Diagnostic Method)

Purpose: To identify slow-moving medicines and evaluate how long inventory remains unsold.

Method: Stock age was calculated as the number of months between the purchase date of each medicine and a reference date (November 2025). Pivot Tables were used to compute average stock age and summarize results for comparison.

Explanation: This analysis helped detect medicines with extended holding periods, indicating slow movement or overstocking. Products with higher stock age were identified as candidates for corrective actions such as reduced reordering or promotional sales.

2.4.5 Analysis of Non-Prescription (OTC) Sales

Purpose: To improve accountability and transparency in non-prescription medicine sales.

Method: Sales records were reviewed to identify non-prescription medicines, and a basic comparison of prescription and non-prescription sales was conducted using Excel summaries.

Explanation: Although detailed documentation was limited, this comparative review provided an initial understanding of the contribution of non-prescription sales to overall revenue and highlighted the need for improved record-keeping to strengthen accountability.

3. Results and Findings

This section presents the key observations obtained from applying the analytical methods described in Section 2. All results are based on the cleaned dataset from April to August 2025 and include sales trends, purchase trends, ABC classification outcomes, expiry-risk insights, and stock-age observations. Charts and tables have been included wherever necessary for clarity and better interpretation.

3.1 Monthly Sales Trend (April–August 2025)

A Pivot Chart was created using monthly sales totals to understand the store's revenue performance over the five-month period. As shown in **Figure 1**, the trend revealed significant variation across months, with the highest sales recorded in **May**, indicating increased customer demand during that period. The lowest sales were observed in **April**, likely due to slower initial movement or buildup of stock.

Sales values for June, July, and August show a stable pattern with mild fluctuations, reflecting consistent monthly demand for commonly purchased items.

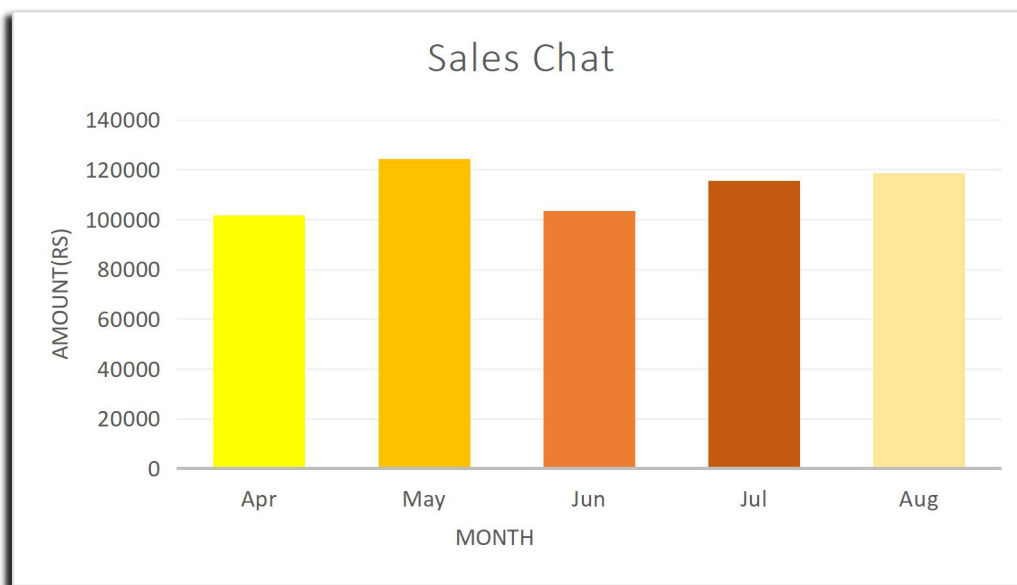


Figure 1: Monthly Sales Trend of Rahul Medical Store (April–August 2025)

3.2 Monthly Purchase Trend (April–August 2025)

As shown in **Figure 2**, the purchase trend indicates that the highest procurement occurred in **April**, which aligns with the beginning of the financial period when stores commonly stock up heavily. The lowest purchase value was observed in **June**, indicating minimal restocking activity during that month.

Purchases in July and August rose moderately, reflecting targeted restocking based on product movement and demand.

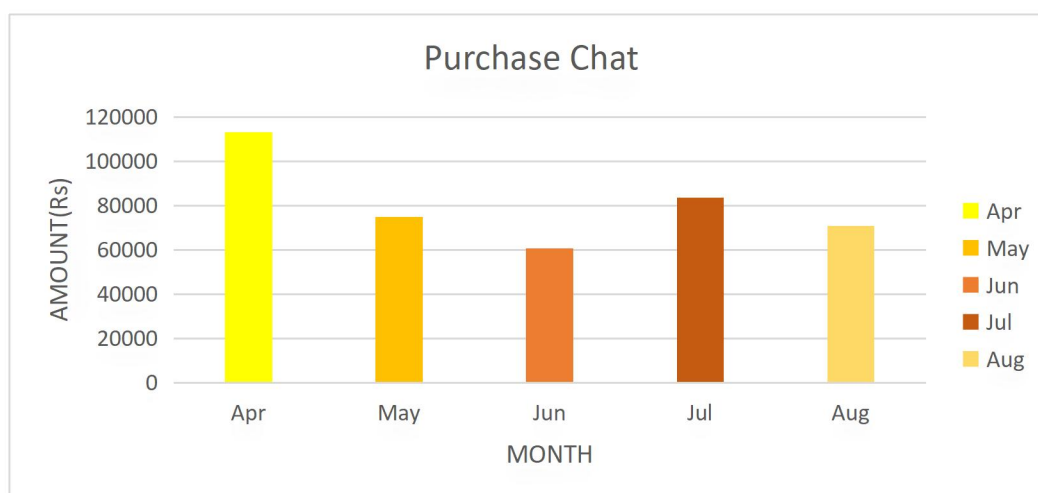


Figure 2: Monthly Purchase Trend of Rahul Medical Store (April–August 2025)

3.3 Combined Sales and Purchase Trend

As shown in **Figure 3**, a combined chart representing both sales and purchases for each month shows the alignment between procurement and revenue generation. Purchases peaked in April, followed by peak sales in May, demonstrating efficient usage of procured inventory.

From June to August, sales and purchases followed a relatively synchronized trend, indicating controlled procurement practices and stable customer demand.

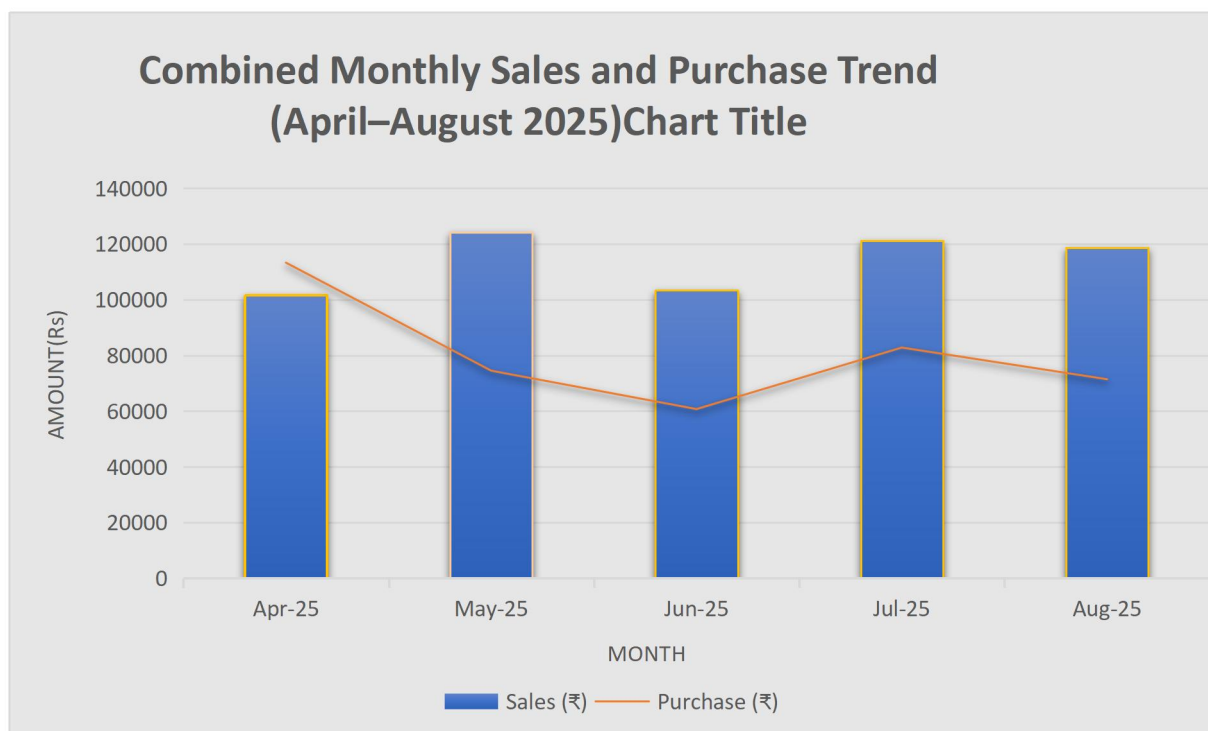


Figure 3: Combined Monthly Sales and Purchase Trend (April–August 2025)

Rahul Medical Store's combined sales and purchase trend from April to August 2025 demonstrates the connection between sales and procurement performance. Sales peaked in May, demonstrating effective use of this inventory, while purchases peaked in April, reflecting the procurement of bulk stock at the start of the fiscal period. Sales and purchases showed a balanced pattern from June to August, indicating steady demand and managed replenishment. All things considered, the two trends' synchronization shows good inventory control and planning.

3.4 ABC Classification Findings

ABC analysis was used to categorize medicines based on their contribution to total purchase value.

Category A (41%) includes high-value medicines that contribute the most to expenditure and require strict monitoring.

Category B (31%) consists of moderately priced items with medium movement.

Category C (28%) contains low-value or slow-moving medicines that pose financial risk due to potential accumulation.

This classification helps the store prioritize stock management efforts and optimize purchase planning.

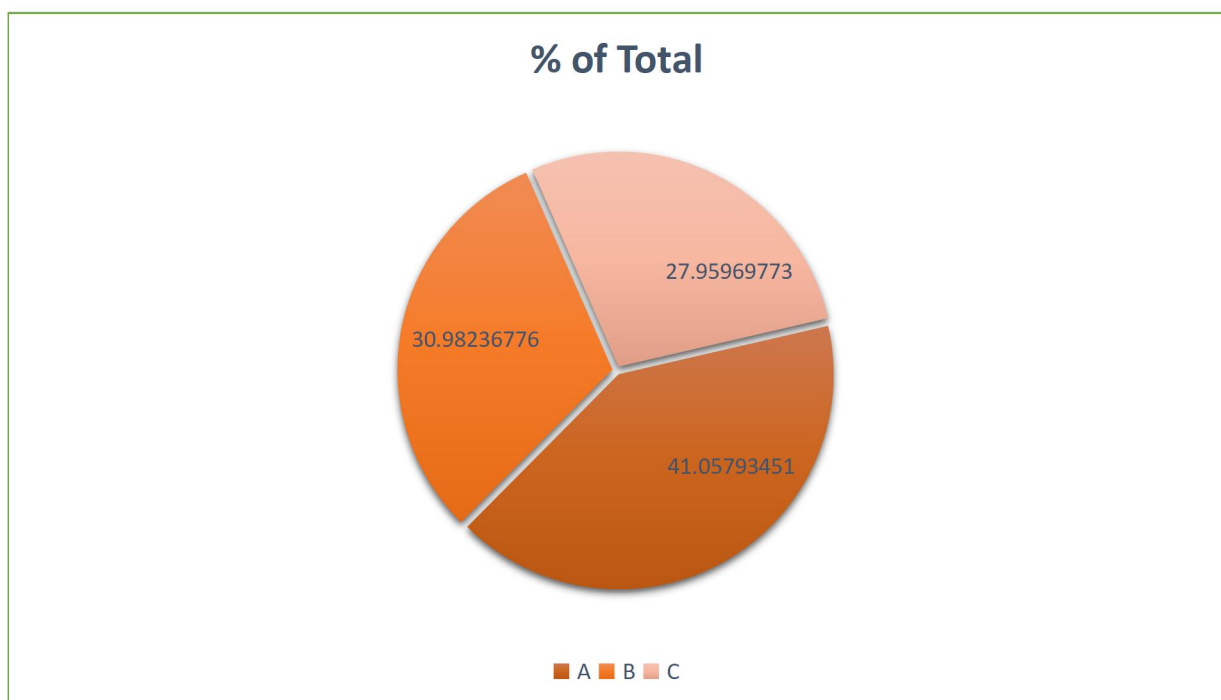


Figure 4: ABC Classification of Medicines at Rahul Medical Store (April–August 2025)

The ABC analysis in **Figure 4** categorizes medicines based on their total purchase value from April to August 2025. As shown, Category A items account for approximately 41% of all medicines, representing the high-value products that contribute the most to total purchase expenditure. Category B covers around 31%, indicating moderately priced or medium-movement items, while Category C represents the remaining 28%, consisting of low-value or slow-moving medicines.

This classification highlights that Rahul Medical Store should closely monitor A-category medicines to ensure optimal stock levels and avoid overinvestment, while B and C items can be managed with routine restocking. Overall, the analysis supports efficient inventory control and prioritization of critical stock items.

3.5 Expiry Risk Analysis

This analysis directly addresses the problem of medicine expiry by identifying products that require immediate action to reduce wastage.

To assess the shelf-life status of the medications in the purchase register, an expiry risk analysis was carried out in addition to value-based classification. Using a date-based condition, the "Expiry Status" column automatically categorized each medication as either Safe or At Risk. In this study, medications that expire between August 2025 and January 2026 were classified as "At Risk," and medications that expire after January 2026 were classified as "Safe."

Only a small percentage of the inventory is approaching expiration, according to the analysis, which suggests efficient stock rotation and prompt procurement. The majority of products are classified as safe, guaranteeing a long enough shelf life for ongoing use and sale. To reduce possible losses from expiration, the At-Risk medications—mostly slow-moving or short-shelf-life items—should be regularly checked, marked down, or sent back to suppliers.

The Expiry Risk Chart (**Figure 5**) visualizes this distribution, highlighting that the majority of the store's stock remains stable, with a minor percentage requiring urgent attention to prevent wastage.

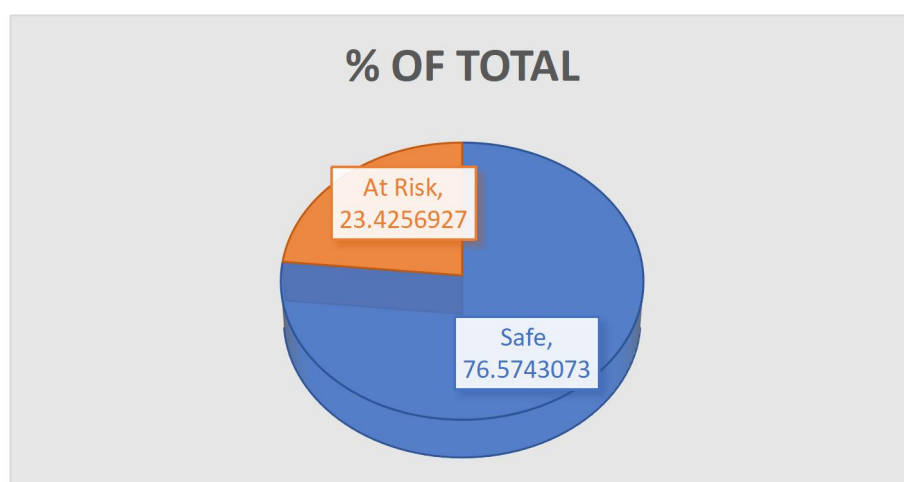


Figure 5: *Expiry Status Distribution of Medicines (Safe vs. At-Risk)*

3.6 Top At-Risk Medicines (By Total Value)

Rank	Product Name	Supplier	Expiry Date	Total Value (₹)	Remarks
1	Volini Gel	Odisha Chemist & Drugist	26-Jan	₹27,295.00	High-value stock nearing expiry; multiple batches
2	Multivitamin Syrup	Popular Medical Store	25-Dec	₹20,531.16	Moderate-value item; slow movement risk
3	Insulin 30/70	Popular Medical Store	26-Jan	₹19,995.50	Essential medicine with moderate expiry exposure

Table 3: Top 3 High-Value Medicines Approaching Expiry (April–August 2025)

The above table identifies the top three medicines contributing the most to the store's expiry risk value. Volini Gel, sourced from Odisha Chemist & Drugist, represents the highest potential loss of ₹27,295 due to multiple batches nearing expiry in January 2026. Multivitamin Syrup and Insulin 30/70, supplied by Popular Medical Store, add another ₹20,531 and ₹19,995 respectively, highlighting significant stock tied up in near-expiry items. These products together constitute a major portion of the at-risk inventory. To mitigate this risk, Rahul Medical Store should prioritize sales of these medicines through discounts, promotional offers, or supplier returns to prevent financial losses.

Overall, the expiry analysis provides valuable insight into how capital gets locked in due to unsold or soon-to-expire products. Addressing these issues early can help the store maintain healthier inventory cycles. The next section, Stock Age and Movement Analysis, further explores how effectively the store's inventory is rotating and identifies areas where product movement can be improved.

3.7 Stock Age Findings

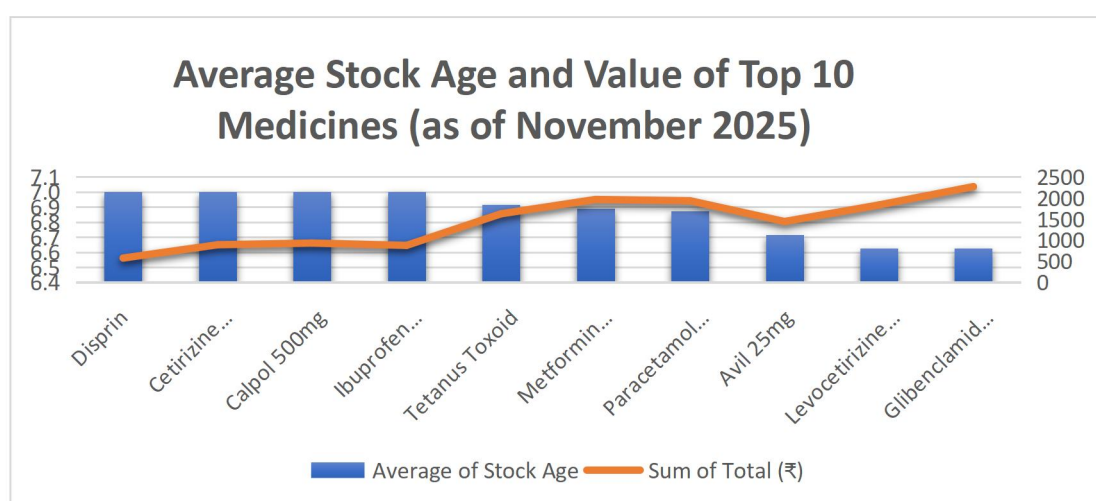


Figure 6: Average Stock Age and Value of Top 10 Medicines (as of November 2025)

As illustrated in **Figure 6**, Disprin, 500 mg of Calpol, 400 mg of Ibuprofen, and 10 mg of Cetirizine have been in stock for almost seven months, which suggests poor movement and a greater chance of stagnation. Despite having comparatively low total values, these products' extended storage can affect how much shelf space is used. Conversely, medications with shorter stock ages—like Avil 25 mg and Paracetamol 500 mg—show faster rotation and steady sales.

In order to maintain the best possible balance between demand and stock availability, Rahul Medical Store can use the combined analysis of stock age and value to identify slow-moving medications, schedule promotional sales, and modify reorder frequencies.

3.8 Prescription and Non-Prescription Medicine Sales Analysis

A comparative analysis was conducted to examine prescription and non-prescription medicine sales using the store's sales register data. Medicines were classified into prescription and non-prescription categories based on common retail dispensing practices, as individual prescription details were not recorded in the dataset.

As shown in **Figure 7**, prescription medicines account for a larger share of total sales value, amounting to ₹4.29 lakh across 1,030 transactions. Non-prescription medicines contribute ₹1.34 lakh through 138 transactions. While non-prescription sales occur less frequently, they still form a meaningful portion of the store's overall revenue.

Unlike prescription medicines, non-prescription transactions are typically recorded without additional verification or supporting documentation. This reduces traceability and accountability, particularly for medicines that may carry misuse or regulatory risks. The findings indicate that even simple documentation or monitoring practices for selected non-prescription medicines could improve transparency and responsible sales management

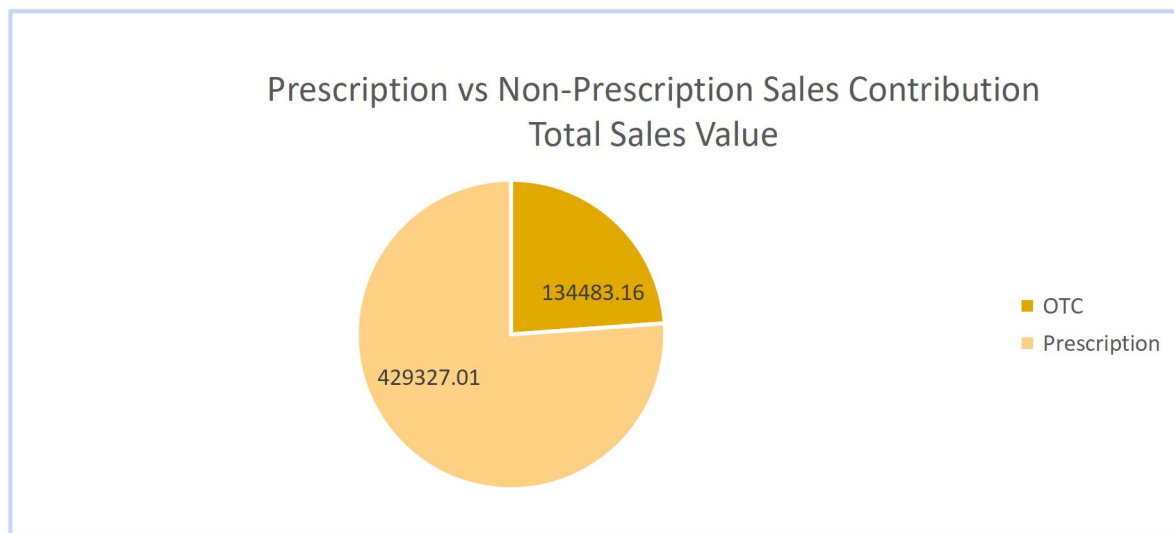


Figure 7: Prescription vs Non-Prescription Sales

4. Interpretation of Results and Recommendations

The analysis conducted using sales data, purchase data, ABC classification, expiry-risk categorization, and stock-age evaluation revealed several important insights about the operational performance of Rahul Medical Store. These insights help explain how the store currently manages its inventory and where improvements are necessary to reduce wastage, avoid financial losses, and ensure more reliable stock availability. This section interprets the key findings and provides practical recommendations that directly address the store's business problems.

4.1 Interpretation of Results

The sales and purchase trends indicate that the store's procurement strategy is front-loaded, with heavy purchases made in April and the highest sales recorded in May. This shows that the store relies on an initial bulk procurement cycle, which is then utilized over the next few months. While this approach ensures that demand is met, it also increases the risk of overstocking, especially for products with limited shelf life. The stable sales volumes from June to August suggest consistent customer demand, but occasional fluctuations highlight the need for more precise forecasting.

The ABC classification revealed that a small number of medicines (Category A) account for a large share of total purchase value. These high-value medicines require the most attention because unsold stock in this category results in the maximum financial loss. Meanwhile, Category C items, though low-value, tend to be slow-moving and can accumulate over time. Effective monitoring of C-category items is important to avoid unnecessary capital lock-in.

The expiry-risk analysis showed that while most medicines fall under the Safe category, a notable portion is set to expire between August 2025 and January 2026. Some of these products also appear in the slow-moving stock category, making them particularly vulnerable to wastage. The presence of at-risk items indicates the need for more systematic rotation of stock and timely actions such as discounts or supplier returns.

The stock-age analysis further revealed that certain products remain in inventory for extended periods, suggesting slow movement or mismatch between customer demand and procurement. Products with high stock age are more likely to become non-moving items or shift into the near-expiry category over time. This emphasizes the need for targeted stock management practices.

The comparison between prescription and non-prescription sales shows that most of the store's revenue comes from prescription medicines, while non-prescription medicines contribute a smaller but noticeable share. However, unlike prescription sales, non-prescription transactions are recorded with limited detail, which reduces traceability. This does not reflect a sales issue, but an accountability gap, indicating the need for simple monitoring practices to support responsible dispensing.

4.2 Recommendations

Based on the insights derived from the analysis, the following recommendations are proposed to improve the store's inventory management, reduce wastage, and support data-driven decision-making:

1. Implement a Structured Reorder Strategy

Instead of relying on bulk procurement at the start of the cycle, the store should consider a staggered purchasing approach based on monthly sales patterns. Category A items should be restocked more frequently in smaller quantities to prevent overstocking, while Category B and C items should be purchased only when required.

2. Prioritize Clearance of At-Risk and Slow-Moving Stock

Medicines identified as "At Risk" should be sold at discounted prices, bundled with frequently purchased items, or returned to suppliers when possible. Slow-moving Category C items should also be reviewed regularly to avoid unnecessary accumulation.

3. Maintain Monthly Inventory Reviews Using Excel

The store already has transaction data in Excel. A simple monthly review using pivot tables can help track sales trends, identify fast- and slow-moving products, and update expiry-risk calculations. This routine will prevent issues such as unexpected stockouts or unnoticed expiry accumulation.

4. Strengthen Documentation for Non-Prescription Sales

Although documentation is minimal for OTC sales, maintaining a simple register for high-risk non-prescription medicines (such as antibiotics or insulin) will improve accountability and help the store stay compliant with regulatory guidelines.

5. Improve Stock Rotation Practices (FIFO Method)

A clear First-In-First-Out (FIFO) approach should be enforced, especially for Category A and B items. Older batches must always be placed at the front of the shelf to ensure that medicines closer to expiry are sold first.

6. Use Visual Expiry Flags in Storage

Color-coded stickers (e.g., Red for <3 months, Yellow for <6 months) can be applied to products nearing expiry. This low-cost method ensures that staff can quickly identify items that require immediate attention, reducing expiry-related losses.

7. Separate Storage for Short Shelf-Life Products

Certain medicines such as syrups, insulin, and ointments have shorter shelf lives. These should be placed in a separate “priority stock” section to avoid neglect during daily store operations.

These recommendations collectively aim to improve stock visibility, enhance purchasing efficiency, and reduce financial loss due to expiry or overstocking. By adopting these simple and practical practices, Rahul Medical Store can transition from an experience-based inventory system to a more reliable, data-driven model.

5. Conclusion

The analysis of Rahul Medical Store’s sales and purchase data from April to August 2025 highlights how inventory decisions, expiry timelines, and product movement influence the store’s operational efficiency. While the store maintains stable sales trends and adequate stock levels, several issues such as overstocking, slow-moving products, and near-expiry medicines continue to create financial risk. By applying Excel-based diagnostic tools such as ABC classification, expiry-risk categorization, and stock-age measurement, the study revealed which products demand immediate attention and which items significantly impact expenditure.

The findings clearly show that data-driven processes can greatly enhance the decision-making quality of small community pharmacies. Implementing structured purchase planning, enforcing FIFO practices, improving documentation, and conducting regular monthly reviews can reduce wastage and improve profitability. This project demonstrates that even simple analytical tools, when applied thoughtfully, can strengthen inventory monitoring and support more sustainable business operations.

6. Future Scope

Although this project provides a foundational analysis of Rahul Medical Store's inventory, there are several opportunities to expand the scope of this work:

- **Integration of a Digital Inventory System:**
A simple software or mobile app can automate stock updates, expiry alerts, and reorder notifications, reducing manual errors.
- **Incorporating Sales Forecasting Models:**
Future analysis can include forecasting (using Excel regression or Python), helping predict demand for seasonal or fast-moving medicines.
- **Supplier Performance Analysis:**
Evaluating supplier consistency, delivery time, pricing, and reliability can help optimize procurement decisions.
- **Margin and Profitability Analysis:**
Including profit margins for each product would allow the store to identify high-profit vs. low-profit medicines.
- **Customer Purchase Pattern Analysis:**
Understanding frequent customer needs could allow the store to stock more effectively and introduce targeted promotional discounts.
- **Automated Expiry Alerts:**
A simple script or Excel macro can provide monthly expiry notifications for items within a 3- or 6-month window.

Overall, these extensions would help build an even more robust and modern inventory management system.

7. Appendix

Appendix A: Figure–Chart Mapping

Figure 1 – Sales_Trend (Monthly Sales Trend (Excel Pivot Chart))

Figure 2 – Purchase_Trend (Monthly Purchase Trend (Excel Pivot Chart))

Figure 3 – Combined_Trend (Combined Monthly Sales and Purchase Trend)

Figure 4 – ABC_Analysis (ABC Classification of Medicines)

Figure 5 – Helper_Stock_Age (Expiry Status Distribution (Safe vs At Risk))

Figure 6 – Stock Age Analysis (Stock Age Analysis of Medicines)

Figure 7 – OTC_vs_Prescription_Analysis (Prescription vs Non-Prescription Sales Comparison)

7.1 Sample Raw Data (Before Cleaning)

RAHUL MEDICAL STORE, Sambalpur, Odisha									
Sales Register									
Period:- April 25 - August 25									
Date	Product Name	Batch No	Expiry Date	Quantity	MRP (₹)	GST (%)	GST Amount (₹)	Amount (₹)	
025-04-01	Disprin	190	May-26	4	5.64	12	2.42	22.56	
025-04-01	Digene	465	Mar-26	3	29.2	12	9.39	87.6	
025-04-01	Multivitamin Syrup	892	Dec-25	3	170	12	54.64	510	
025-04-01	Ibuprofen 400mg	222	May-26	6	12.88	12	8.28	77.28	
025-04-01	Erythromycin 250mg	377	Apr-26	7	60	12	45.00	420	
025-04-01	Tetanus Toxoid	981	Apr-28	3	11.35	12	3.65	34.05	
025-04-01	Amlodipine 5mg	410	Dec-26	2	83.9	12	17.98	167.8	
025-04-02	Tetanus Toxoid	981	Apr-28	5	11.35	12	6.08	56.75	
025-04-02	Volini Gel	531	Jan-26	4	265	12	113.57	1060	
025-04-02	Azithromycin 500mg	885	Jul-26	2	79.16	12	16.96	158.32	
025-04-02	Hand Sanitizer	839	Jan-26	7	30	12	22.50	210	
025-04-02	Hydroxychloroquine 200mg	963	Apr-26	2	108.7	12	23.29	217.4	
025-04-02	Loperamide 2mg	751	Apr-26	10	25.48	12	27.30	254.8	

- This snapshot represents the raw data as recorded in the physical registers before cleaning and standardization.

7.2 Cleaned Data Format

RAHUL MEDICAL STORE, Sambalpur, Odisha									
Sales Register									
Period:- April 25 - August 25									
Date	Product Name	Batch No	Expiry Date	Quantity	MRP (₹)	GST (%)	GST Amount (₹)	Amount (₹)	Month_Date
2025-04-01	Disprin	190	May-26	4	5.64	12	2.42	22.56	4/1/2025
2025-04-01	Digene	465	Mar-26	3	29.2	12	9.39	87.6	4/1/2025
2025-04-01	Multivitamin Syrup	892	Dec-25	3	170	12	54.64	510	4/1/2025
2025-04-01	Ibuprofen 400mg	222	May-26	6	12.88	12	8.28	77.28	4/1/2025
2025-04-01	Erythromycin 250mg	377	Apr-26	7	60	12	45.00	420	4/1/2025
2025-04-01	Tetanus Toxoid	981	Apr-28	3	11.35	12	3.65	34.05	4/1/2025
2025-04-01	Amlodipine 5mg	410	Dec-26	2	83.9	12	17.98	167.8	4/1/2025
2025-04-02	Tetanus Toxoid	981	Apr-28	5	11.35	12	6.08	56.75	4/1/2025
2025-04-02	Volini Gel	531	Jan-26	4	265	12	113.57	1060	4/1/2025
2025-04-02	Azithromycin 500mg	885	Jul-26	2	79.16	12	16.96	158.32	4/1/2025
2025-04-02	Hand Sanitizer	839	Jan-26	7	30	12	22.50	210	4/1/2025
2025-04-02	Hydroxychloroquine 200mg	963	Apr-26	2	108.7	12	23.29	217.4	4/1/2025
2025-04-02	Loperamide 2mg	751	Apr-26	10	25.48	12	27.30	254.8	4/1/2025
2025-04-02	Clotrimazole Cream	440	Apr-26	4	302.5	12	129.64	1210	4/1/2025
2025-04-02	Zincovit	734	Mar-26	2	115	12	24.64	230	4/1/2025
2025-04-03	Montelukast 10mg	648	Jan-27	2	140	12	30.00	280	4/1/2025

- This dataset was used as the master dataset for all analyses including sales trends, ABC classification, expiry risk, and stock age analysis.

7.3 ABC Classification Calculation Snapshot

=IF(O5<=0.8,"A",IF(O5<=0.95,"B","C"))

- ABC classification was performed by sorting medicines in descending order of total purchase value and computing cumulative percentages to assign A, B, and C categories.

7.4 Expiry-Risk Formula Used

L5											
=IF(AND(E5>=DATE(2025,8,1),E5<=DATE(2026,1,31)),"At Risk","Safe")											
	A	B	C	D	E	F	G	H	I	J	K
3	Period:- April 25 - August25										
4	Site	Owner/Supplier	Product Name	Batch No	Expiry Date	Quantity	Rate (₹)	Amount (₹)	GST (%)	T Amount	Total (₹)
5	5-04-2	Odisha Ch	Volini Gel	144	Jan-26	60	218.36	11697.86	12	1403.74	13101.60
6	5-08-1	Taj Pharm	Clotrimaz	522	May-26	30	187.55	5023.66	12	602.84	5626.50
7	5-08-1	Taj Pharm	Clotrimaz	522	May-26	30	187.55	5023.66	12	602.84	5626.50
8	5-04-0	Odisha Ch	Volini Gel	994	Jan-26	25	218.36	4874.11	12	584.89	5459.00
											Expiry Status

Medicines expiring between August 2025 and January 2026 were classified as “At Risk”.

7.5 Stock Age Formula Used

M5											
=DATEDIF(A5,DATE(2025,11,30),"m")											
	A	B	C	D	E	F	G	H	I	J	K
1	RAHUL MEDICAL STORE,Sambalpur,Odisha										
2	Purchase Register										
3	Period:- April 25 - August25										
4	Date	Owner/Supplier	Product Name	Batch No	Expiry Date	Quantity	Rate (₹)	Amount (₹)	GST (%)	T Amount	Total (₹)
5	2025-04-01	Utkal Phar	Disprin	292	May-26	10	4.51	40.27	12	4.83	45.10
6	2025-04-01	Utkal Phar	Disprin	598	May-26	10	4.51	40.27	12	4.83	45.10
7	2025-04-01	Utkal Phar	Disprin	736	May-26	10	4.51	40.27	12	4.83	45.10
8	2025-04-01	Utkal Phar	Disprin	736	May-26	10	4.51	40.27	12	4.83	45.10
9	2025-04-01	Utkal Phar	Disprin	414	May-26	15	4.51	60.40	12	7.25	67.65
10	2025-04-01	Utkal Phar	Disprin	521	May-26	15	4.51	60.40	12	7.25	67.65
11	2025-04-02	Utkal Phar	Disprin	521	May-26	15	4.51	60.40	12	7.25	67.65
12	2025-04-02	Ashok Pha	ORS Sachet	903	Dec-25	5	14.73	65.76	12	7.89	73.65
											Expiry Status
											Stock Age

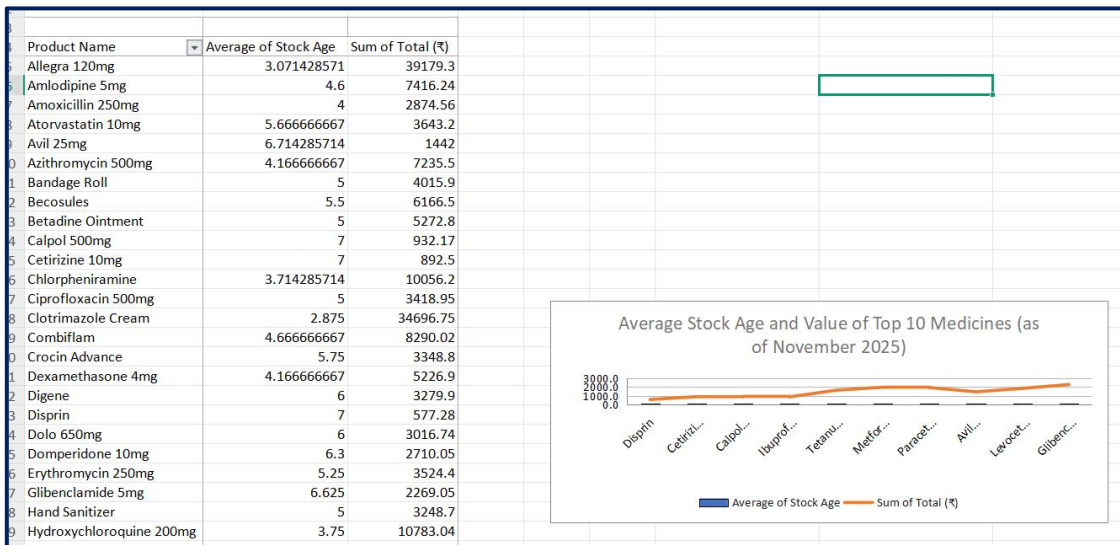
- Stock age was calculated as the number of months between the purchase date and the reference date (November 2025).

7.6 Prescription vs Non-Prescription Sales Pivot Table

Row Labels	Total Sales Value (₹)	Number of Transactions
OTC	134483.16	138
Prescription	429327.01	1030
Grand Total	563810.17	1168

- This pivot table was used to compare prescription and non-prescription medicine sales for accountability analysis.

7.7 Other Screenshots



H21													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Date	Custom	Product	Batch N	Expiry D	Quantit	Rate (₹)	Amount (₹)	GST (%)	GST Amount	Total (₹)	Expiry S	Stock A
2	2025-08-0	Popular M	Allegra 120	830	7/1/2026	8	211.78	1512.714286	12	181.5257143	1694.24	Safe	3
3	2025-08-0	Popular M	Allegra 120	104	7/1/2026	10	211.78	1890.892857	12	226.9071429	2117.8	Safe	3
4	2025-08-0	Popular M	Allegra 120	356	7/1/2026	10	211.78	1890.892857	12	226.9071429	2117.8	Safe	3
5	2025-08-0	Popular M	Allegra 120	175	7/1/2026	10	211.78	1890.892857	12	226.9071429	2117.8	Safe	3
6	2025-08-0	Popular M	Allegra 120	289	1/1/2027	10	211.78	1890.892857	12	226.9071429	2117.8	Safe	3
7	2025-08-0	Popular M	Allegra 120	456	7/1/2026	10	211.78	1890.892857	12	226.9071429	2117.8	Safe	3
8	2025-08-1	Popular M	Allegra 120	171	7/1/2026	11	211.78	2079.982143	12	249.5978571	2329.58	Safe	3
9	2025-08-1	Popular M	Allegra 120	271	7/1/2026	11	211.78	2079.982143	12	249.5978571	2329.58	Safe	3
10	2025-07-2	Popular M	Allegra 120	872	1/1/2027	15	211.78	2836.339286	12	340.3607143	3176.7	Safe	4
11	2025-08-2	Popular M	Allegra 120	394	1/1/2027	15	211.78	2836.339286	12	340.3607143	3176.7	Safe	3
12	2025-08-2	Popular M	Allegra 120	672	1/1/2027	15	211.78	2836.339286	12	340.3607143	3176.7	Safe	3
13	2025-08-2	Popular M	Allegra 120	294	1/1/2027	15	211.78	2836.339286	12	340.3607143	3176.7	Safe	3
14	2025-08-2	Popular M	Allegra 120	782	1/1/2027	20	211.78	3781.785714	12	453.8142857	4235.6	Safe	3
15	2025-08-2	Popular M	Allegra 120	682	1/1/2027	25	211.78	4727.232143	12	567.2678571	5294.5	Safe	3

Month	Total Sales (₹)	Total Purchases (₹)	Observation
25-Apr	1,01,604.5	1,13,276.8	Initial stock buildup
25-May	1,24,272.3	74,483.70	Peak sales month
25-Jun	1,03,507.5	60,699.40	Lowest purchase month
25-Jul	1,21,271.1	82,787.50	Demand stable
25-Aug	1,18,691.1	71,409.90	Moderate restocking

Category	Description	% of Total Value	Inventory Control Priority
A	High-value and fast-moving medicines	41%	Very High
B	Moderately priced and medium-movement items	31%	Medium
C	Low-value, slow-moving or infrequent sales items	28%	Low

Row Labels	Average of Stock Age	Sum of Total (₹)							
Disprin	7.0	577.28							
Cetirizine 10mg	7.0	892.5							
Calpol 500mg	7.0	932.17							
Ibuprofen 400mg	7.0	877.45							
Tetanus Toxoid	6.9	1626							
Metformin 500mg	6.9	1964.6							
Paracetamol 500mg	6.9	1934.4							
Avil 25mg	6.7	1442							
Levocetirizine 5mg	6.6	1834.35							
Glibenclamide 5mg	6.6	2269.05							
Ranitidine 150mg	6.6	1279.5							
ORS Sachet	6.5	1841.25							
Omeprazole 20mg	6.3	1814.4							
Domperidone 10mg	6.3	2710.05							
Prednisolone 10mg	6.2	1575.99							
Dolo 650mg	6.0	3016.74							
Digene	6.0	3279.9							
Vitamin C Tablets	6.0	2353.5							
N95 Mask	5.8	2479.1							
Loperamide 2mg	5.8	2848.9							
CrocIn Advance	5.8	3348.8							
Atorvastatin 10mg	5.7	3643.2							

Products	Average of Stock Age	Sum of Total (₹)
Disprin	7.0	577.28
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Calpol 500mg	7.0	932.17
Ibuprofen	7.0	877.45
Tetanus Toxoid	6.9	1626
Metformin	6.9	1964.6
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Avil 25mg	6.7	1442
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Glibenclamide	6.6	2269.05